Model Comparison MAE Table

Setup

```
packages <- c("readxl", "forecast", "dlnm", "splines", "Metrics",
              "demography", "tseries", "mgcv", "dplyr", "gridExtra",
              "writexl", "hwwntest", "ISOweek", "seastests", "MTS",
              "tidyr", "grid", "nortest")
invisible(lapply(packages, library, character.only = TRUE))

find_project_root <- function(start = getwd()) {
  path <- normalizePath(start, mustWork = FALSE)
  for (i in seq_len(8)) {
    if (dir.exists(file.path(path, "Code")) && dir.exists(file.path(path, "Data"))) return(path)
    parent <- dirname(path)
    if (identical(parent, path)) break
    path <- parent
  }
  stop("Set LC_DLNM_BASE_DIR to the project root.")
}
base_dir <- Sys.getenv("LC_DLNM_BASE_DIR")
if (!nzchar(base_dir)) base_dir <- find_project_root()
data_dir <- file.path(base_dir, "Data/Combined_data")
function_dir <- file.path(base_dir, "Code/Function")
results_dir <- file.path(base_dir, "Results/Model-comparison")
dir.create(results_dir, recursive = TRUE, showWarnings = FALSE)

files <- list.files(function_dir, pattern = "\\.R$", full.names = TRUE)
invisible(sapply(files, source))

sets <- 10
size <- 8
forecast_step <- 78
start_ind <- 102
lag_days <- 21
var_df <- 3
lag_df <- 3
spline_degree <- 3
season_df <- 4
max_iter_ll <- 20

regions <- c("Athens", "Lisbon", "Rome")
model_names <- c("LC", "LL", "DLNM--LC", "DLNM--LL", "Madaniyazi", "Guibert")
age_rows <- c("20--64", "65--74", "75--84", "85+")
age_cols <- c("Y20_64", "Y65_74", "Y75_84", "Y_GE85")
age_groups <- c("20-64", "65-74", "75-84", "85+")
```

```r
region_config <- list(
  Attiki = list(display = "Athens", death_col = 3, exposure_col = 9,
                max_iter = 20),
  Lisbon = list(display = "Lisbon", death_col = 4, exposure_col = 10,
                max_iter = 20),
  Roma = list(display = "Rome", death_col = 6, exposure_col = 12,
              max_iter = 15)
)

Y20_64 <- read_xlsx(file.path(data_dir, "Y20_64_combined.xlsx"))
Y65_74 <- read_xlsx(file.path(data_dir, "Y65_74_combined.xlsx"))
Y75_84 <- read_xlsx(file.path(data_dir, "Y75_84_combined.xlsx"))
Y_GE85 <- read_xlsx(file.path(data_dir, "Y_GE85_combined.xlsx"))

region_data <- list()
for (region in names(region_config)) {
  cfg <- region_config[[region]]

  death <- cbind(Y20_64[[cfg$death_col]],
                 Y65_74[[cfg$death_col]],
                 Y75_84[[cfg$death_col]],
                 Y_GE85[[cfg$death_col]])
  exposure <- cbind(Y20_64[[cfg$exposure_col]],
                    Y65_74[[cfg$exposure_col]],
                    Y75_84[[cfg$exposure_col]],
                    Y_GE85[[cfg$exposure_col]])
  rate <- death / exposure * 52
  colnames(death) <- colnames(exposure) <- colnames(rate) <- age_cols
  rownames(death) <- rownames(exposure) <- rownames(rate) <- Y20_64$Week

  wave <- data.frame(
    hot_wave = Y20_64[[paste0(region, "_hot_wave3")]],
    cold_wave = Y20_64[[paste0(region, "_cold_wave3")]]
  )
  colnames(wave) <- c(paste0(region, "_hot_wave3"),
                      paste0(region, "_cold_wave3"))

  region_data[[region]] <- list(
    display = cfg$display,
    max_iter = cfg$max_iter,
    death = death,
    exposure = exposure,
    rate = rate,
    wave = wave,
    utci = load_crossbasis_history(base_dir, cfg$display, region)
  )
}

cv_folds <- lapply(0:(sets - 1), function(i) {
  train_end <- start_ind + i * size
  test_start <- train_end + 1
  test_end <- train_end + forecast_step
  if (test_end > nrow(Y20_64)) stop("Forecast window exceeds available data.")
```

```
  list(train = 1:train_end, test = test_start:test_end)
})
```

Expanding-Window Cross-Validation

LC and DLNM–LC

```
lc_rows <- list()
dlnm_lc_rows <- list()

for (region in names(region_config)) {
  dat <- region_data[[region]]

  fold_mae <- lapply(cv_folds, function(fold) {
    utci_train <- dat$utci[fold$train, ]
    utci_test <- dat$utci[fold$test, ]
    varknots <- equalknots(utci_train, fun = "ns", df = var_df,
                           degree = spline_degree)
    lagknots <- logknots(lag_days, lag_df)

    cb_train <- crossbasis(
      utci_train, lag = lag_days,
      argvar = list(fun = "ns", knots = varknots),
      arglag = list(knots = lagknots, df = lag_df)
    )
    cb_test <- crossbasis(
      utci_test, lag = lag_days,
      argvar = list(fun = "ns", knots = varknots),
      arglag = list(knots = lagknots, df = lag_df)
    )

    train_model <- DLNM_LC(dat$rate[fold$train, ],
                           cb_train,
                           dat$wave[fold$train, ],
                           tol = 1e-2,
                           max_iter = dat$max_iter,
                           region = region)
    fc <- DLNM_LC.forecast(
      train_model,
      cb_test,
      dat$wave[fold$test, ],
      kt_model_type = "SARIMA",
      forecast_step = forecast_step
    )

    rbind(
      LC = 100 * sapply(seq_along(age_cols), function(j) {
        mae(dat$rate[fold$test, j], fc$baseline_LC.forecast[, j])
      }),
      `DLNM--LC` = 100 * sapply(seq_along(age_cols), function(j) {
        mae(dat$rate[fold$test, j],
            fc$DLNM_LC_final.forecast$model.forecast[, j])
```

```r
      })
    )
  })

  mean_mae <- Reduce(`+`, fold_mae) / length(fold_mae)
  colnames(mean_mae) <- age_rows
  lc_rows[[dat$display]] <- mean_mae["LC", ]
  dlnm_lc_rows[[dat$display]] <- mean_mae["DLNM--LC", ]
}

lc_matrix <- do.call(rbind, lc_rows)[regions, age_rows, drop = FALSE]
dlnm_lc_matrix <- do.call(rbind, dlnm_lc_rows)[regions, age_rows, drop = FALSE]
storage.mode(lc_matrix) <- "numeric"
storage.mode(dlnm_lc_matrix) <- "numeric"

lc_result <- list(
  LC = lc_matrix,
  `DLNM--LC` = dlnm_lc_matrix
)
```

LL and DLNM–LL

```r
rate_list <- lapply(region_data, `[[`, "rate")
wave_list <- lapply(region_data, `[[`, "wave")
utci_list <- lapply(region_data, `[[`, "utci")

folds <- lapply(0:(sets - 1), function(i) {
  train_size <- start_ind + i * size
  test_size <- train_size + forecast_step
  create_train_test_sets(rate_list, wave_list, utci_list,
                         train_size = train_size,
                         test_size = test_size,
                         lag = lag_days,
                         var.df = var_df,
                         lag.df = lag_df,
                         degree = spline_degree)
})

fold_mae <- lapply(folds, function(fold) {
  train_model <- DLNM_LL(dat_list = fold$train_set$dat_list,
                         cb_list = fold$train_set$cb_list,
                         wave_list = fold$train_set$wave_list,
                         tol = 1e-2,
                         max_iter = max_iter_ll)
  fc <- DLNM_LL.forecast(
    train_model,
    cb_matrix_test = fold$test_set$cb_list,
    wave_data_test = fold$test_set$wave_list,
    forecast_step = nrow(fold$test_set$dat_list[[1]])
  )

  baseline_ll <- lapply(fc$baseline_LL.forecast, function(region) {
    as.data.frame(lapply(region, exp))
```

```r
  })
  dlnm_ll <- lapply(fc$DLNM_LL_final.forecast$model.forecast, function(region) {
    as.data.frame(lapply(region, exp))
  })

  lapply(names(fold$test_set$dat_list), function(region) {
    test_rate <- fold$test_set$dat_list[[region]]
    rbind(
      LL = 100 * sapply(seq_along(age_cols), function(j) {
        mae(test_rate[, j], baseline_ll[[region]][, j])
      }),
      `DLNM--LL` = 100 * sapply(seq_along(age_cols), function(j) {
        mae(test_rate[, j], dlnm_ll[[region]][, j])
      })
    )
  }) |>
    setNames(names(fold$test_set$dat_list))
})

ll_rows <- list()
dlnm_ll_rows <- list()
for (region in names(region_config)) {
  mean_mae <- Reduce(`+`, lapply(fold_mae, function(x) x[[region]])) /
    length(fold_mae)
  colnames(mean_mae) <- age_rows
  display <- region_config[[region]]$display
  ll_rows[[display]] <- mean_mae["LL", ]
  dlnm_ll_rows[[display]] <- mean_mae["DLNM--LL", ]
}

ll_matrix <- do.call(rbind, ll_rows)[regions, age_rows, drop = FALSE]
dlnm_ll_matrix <- do.call(rbind, dlnm_ll_rows)[regions, age_rows, drop = FALSE]
storage.mode(ll_matrix) <- "numeric"
storage.mode(dlnm_ll_matrix) <- "numeric"

ll_result <- list(
  LL = ll_matrix,
  `DLNM--LL` = dlnm_ll_matrix
)
```

Madaniyazi Model

```r
week_all <- Y20_64$Week
weekly_trend <- seq_along(week_all) / 52
week_of_year <- as.integer(sub(".*-W", "", week_all))
season <- mgcv::smoothCon(
  mgcv::s(week_of_year, bs = "cc", k = season_df + 2, fx = TRUE),
  data = data.frame(week_of_year = week_of_year),
  absorb.cons = TRUE
)[[1]]$X

madaniyazi_rows <- list()
for (region in names(region_config)) {
```

```r
  dat <- region_data[[region]]

  madaniyazi_rows[[dat$display]] <- 100 * sapply(seq_along(age_cols), function(j) {
    fold_mae <- sapply(cv_folds, function(fold) {
      utci_train <- dat$utci[fold$train, ]
      utci_test <- dat$utci[fold$test, ]
      varknots <- equalknots(utci_train, fun = "ns", df = var_df,
                             degree = spline_degree)
      lagknots <- logknots(lag_days, fun = "ns", df = lag_df)

      cb_train <- crossbasis(
        utci_train, lag = lag_days,
        argvar = list(fun = "ns", knots = varknots),
        arglag = list(fun = "ns", knots = lagknots, df = lag_df)
      )
      cb_test <- crossbasis(
        utci_test, lag = lag_days,
        argvar = list(fun = "ns", knots = varknots),
        arglag = list(fun = "ns", knots = lagknots, df = lag_df)
      )

      fit <- Madaniyazi.fit(
        death = as.numeric(dat$death[fold$train, j]),
        exposure = as.numeric(dat$exposure[fold$train, j]),
        utci_crossbasis = cb_train,
        weekly_trend = weekly_trend[fold$train],
        week_of_year_basis = season[fold$train, , drop = FALSE]
      )
      pred_rate <- Madaniyazi.forecast(
        fit,
        exposure = as.numeric(dat$exposure[fold$test, j]),
        utci_crossbasis = cb_test,
        weekly_trend = weekly_trend[fold$test],
        week_of_year_basis = season[fold$test, , drop = FALSE]
      )
      mae(dat$rate[fold$test, j], pred_rate)
    })
    mean(fold_mae, na.rm = TRUE)
  })
}

madaniyazi_result <- do.call(rbind, madaniyazi_rows)
colnames(madaniyazi_result) <- age_rows
madaniyazi_result <- madaniyazi_result[regions, age_rows, drop = FALSE]
storage.mode(madaniyazi_result) <- "numeric"
```

Guibert Model

```r
rate_list <- lapply(region_data, function(dat) {
  data.frame(time = seq_len(nrow(Y20_64)), dat$rate)
})
utci_list <- lapply(region_data, `[[`, "utci")
```

```r
folds <- lapply(0:(sets - 1), function(i) {
  train_size <- start_ind + i * size
  test_size <- train_size + forecast_step
  Guibert.create_train_test_sets(rate_list, utci_list,
                                 train_size = train_size,
                                 test_size = test_size,
                                 lag = lag_days,
                                 var.df = var_df,
                                 lag.df = lag_df,
                                 degree = spline_degree)
})

fold_mae <- lapply(folds, function(fold) {
  fit <- Guibert.fit(regions_data = fold$train_set$dat_list,
                     cb_list = fold$train_set$cb_list,
                     ns_df = 8)
  fc <- Guibert.forecast(
    LL_fit = fit$LL_fit,
    dlnm_estimates_list = fit$dlnm_estimates_list,
    cb_matrix_test = fold$test_set$cb_list,
    forecast_step = nrow(fold$test_set$dat_list[[1]])
  )

  lapply(names(fc$Guibert.forecast.list), function(region) {
    forecast_rate <- as.data.frame(fc$Guibert.forecast.list[[region]])
    test_rate <- as.data.frame(fold$test_set$dat_list[[region]])
    100 * sapply(age_cols, function(age) {
      mae(test_rate[[age]], forecast_rate[[age]])
    })
  }) |>
    setNames(names(fc$Guibert.forecast.list))
})

guibert_rows <- list()
for (region in names(region_config)) {
  display <- region_config[[region]]$display
  guibert_rows[[display]] <- rowMeans(do.call(cbind, lapply(fold_mae, function(x) {
    x[[region]][age_cols]
  })), na.rm = TRUE)
}

guibert_result <- do.call(rbind, guibert_rows)
colnames(guibert_result) <- age_rows
guibert_result <- guibert_result[regions, age_rows, drop = FALSE]
storage.mode(guibert_result) <- "numeric"
```

Forecast MAE Summary

```r
model_mae <- list(
  LC = lc_result$LC,
  LL = ll_result$LL,
  `DLNM--LC` = lc_result$`DLNM--LC`,
  `DLNM--LL` = ll_result$`DLNM--LL`,
```

```
  Madaniyazi = madaniyazi_result,
  Guibert = guibert_result
)

model_mae <- model_mae[model_names]

comparison_long <- do.call(rbind, lapply(regions, function(region) {
  do.call(rbind, lapply(age_rows, function(age_group) {
    data.frame(
      Region = region,
      Age_group = age_group,
      t(vapply(model_names, function(model) {
        model_mae[[model]][region, age_group]
      }, numeric(1))),
      check.names = FALSE,
      row.names = NULL
    )
  }))
}))

write.csv(comparison_long,
          file.path(results_dir, "Model-comparison-MAE-table.csv"),
          row.names = FALSE)
```

```
latex_table <- c(
  "\\begin{table}[H]",
  "    \\centering",
  "    \\color{black}",
  "    \\resizebox{\\textwidth}{!}{",
  "    \\begin{tabular}{ccccccc}"
)

for (region in regions) {
  latex_table <- c(
    latex_table,
    if (region == regions[1]) "        \\toprule" else "        \\midrule",
    sprintf("        \\small\\textbf{%s} ", region),
    "          & \\small\\textbf{LC} ",
    "          & \\small\\textbf{LL} ",
    "          & \\small\\textbf{DLNM--LC} ",
    "          & \\small\\textbf{DLNM--LL} ",
    "          & \\small\\textbf{Madaniyazi et al.}",
    "          & \\small\\textbf{Guibert et al.} \\\\",
    "        \\midrule"
  )

  for (age_group in age_rows) {
    values <- round(vapply(model_names, function(model) {
      model_mae[[model]][region, age_group]
    }, numeric(1)), 4)
    distinct_values <- sort(unique(values[is.finite(values)]))
    best_value <- distinct_values[1]
    second_value <- if (length(distinct_values) > 1) distinct_values[2] else NA_real_
```

```r
    cells <- sprintf("%.4f", values)
    cells[values == best_value] <- paste0("\\textbf{",
                                          cells[values == best_value],
                                          "}")
    if (is.finite(second_value)) {
      cells[values == second_value] <- paste0("\\textit{",
                                              cells[values == second_value],
                                              "}")
    }

    latex_table <- c(
      latex_table,
      sprintf("        %s & %s \\\\", age_group, paste(cells, collapse = " & "))
    )
  }
}

latex_table <- c(
  latex_table,
  "         \\bottomrule",
  "    \\end{tabular}}",
  "    \\caption{MAE ($\\times 100$) for 10--fold expanding window cross-validation.}",
  "    \\label{tab:compare_mae}",
  "\\end{table}"
)

writeLines(latex_table,
           file.path(results_dir, "Model-comparison-MAE-table.tex"))
cat(paste(latex_table, collapse = "\n"))
```

Athens	**LC**	**LL**	**DLNM–LC**	**DLNM–LL**	**Madaniyazi et al.**	**Guibert et al.**
20–64	0.0258	0.0285	**0.0231**	*0.0233*	0.0241	0.0298
65–74	0.1636	0.1830	**0.1355**	0.1565	*0.1385*	0.1787
75–84	0.5009	0.5091	*0.3675*	0.3767	**0.3466**	0.4122
85+	1.8120	2.0056	**1.2836**	1.5514	*1.2899*	1.5648
Lisbon	**LC**	**LL**	**DLNM–LC**	**DLNM–LL**	**Madaniyazi et al.**	**Guibert et al.**
20–64	0.0274	0.0274	*0.0257*	*0.0257*	**0.0246**	0.0284
65–74	0.1500	0.1624	0.1592	**0.1382**	0.1523	*0.1404*
75–84	0.4615	0.5012	0.5032	*0.4055*	**0.3869**	0.5085
85+	1.8087	2.0249	1.7293	*1.6425*	**1.4382**	1.7378
Rome	**LC**	**LL**	**DLNM–LC**	**DLNM–LL**	**Madaniyazi et al.**	**Guibert et al.**
20–64	0.0197	*0.0192*	0.0204	0.0206	**0.0185**	0.0235
65–74	**0.1189**	*0.1276*	0.1491	0.1475	0.1346	0.1394
75–84	0.3455	0.3927	0.3511	**0.3254**	*0.3438*	0.3635
85+	1.4920	1.7384	*1.4388*	**1.3179**	1.5267	1.7075

Table 1: MAE ($\times 100$) for 10–fold expanding window cross-validation.